④ The InstanceSecurityGroup resource is not yet specified; you will add this later when you define the EC2 instance that runs the web server.

⑤ The cache subnet group references the VPC subnets.

⑥ Creates a single-node Redis cluster

⑦ You can specify the exact version of Redis that you want to run. Otherwise, the latest version is used, which may cause incompatibility issues in the future. We recommend always specifying the version.

The single-node Redis cache cluster is now defined. Discourse also requires a PostgreSQL database, which you'll define next.

11.4.3 Database: Security group, subnet group, database instance

PostgreSQL is a powerful, open source relational database. If you are not familiar with PostgreSQL, that's not a problem at all. Luckily, the RDS service will provide a managed PostgreSQL database for you. You learned about RDS in chapter 10. The following listing shows the section of the template that defines the RDS instance.

Listing 11.5 CloudFormation template for Discourse: Database

```
Resources:
  # [...]
  DatabaseSecurityGroup:                                          ①
    Type: 'AWS::EC2::SecurityGroup'
    Properties:
      GroupDescription: database
      VpcId: !Ref VPC
  DatabaseSecurityGroupIngress:
    Type: 'AWS::EC2::SecurityGroupIngress'
    Properties:
      GroupId: !Ref DatabaseSecurityGroup
      IpProtocol: tcp
      FromPort: 5432                                              ②
      ToPort: 5432
      SourceSecurityGroupId: !Ref InstanceSecurityGroup           ③
  DatabaseSubnetGroup:                                            ④
    Type: 'AWS::RDS::DBSubnetGroup'
    Properties:
      DBSubnetGroupDescription: database
      SubnetIds:
      - Ref: SubnetA
      - Ref: SubnetB
  Database:                                                       ⑤
    Type: 'AWS::RDS::DBInstance'
    DeletionPolicy: Delete
    Properties:
      AllocatedStorage: 5
      BackupRetentionPeriod: 0                                    ⑥
      DBInstanceClass: 'db.t2.micro'
```